A p-Opportunistic Channel Access Scheme for Interference Mitigation Between V2V and V2I Communications

Xiangming Wen, *Senior Member, IEEE*, Jenhui Chen, *Senior Member, IEEE*, Zhiqun Hu, and Zhaoming Lu

***Abstract*—In this article, we study the co-channel problem in the 2-tier architecture of vehicular networks [i.e., vehicle-to-vehicle (V2V) communication and vehicle-to-infrastructure (V2I) communication]. The communication technology we consider here is dedicated to short-range radio communication (DSRC), cellular vehicle-to-everything (C-V2X), or a hybrid of both. The V2I communication will interfere V2V communication, and *vice versa*, because the roadside unit (RSU) cannot sense V2V communication during the downlink period when the V2V communication is in the coverage of RSU. We note that V2V communication will have higher priority since it conveys critical messages for road safety in connected and automated vehicle (CAV) systems. We propose a p-opportunity channel access scheme (p-OCAS) for the RSU to solve the problem. Simulation results validate the correctness of the analytical model. The investigation showed that p-OCAS can substantially minimize the interference from RSU to V2V communications according to the V2V session arrival rate to automated vehicles as well as maintain a high throughput of RSU.**

***Index Terms*—Automated vehicle (AV), cellular vehicle-to-everything (C-V2X), interference, media access control (MAC), vehicle-to-infrastructure (V2I), vehicle-to-vehicle (V2V).**

I. Introduction

VEHICULAR communication is an indispensable part of accomplishing unmanned vehicular networks in intelligent transportation systems (ITSs) [1], [2]. In vehicular networks, automated vehicles (AVs), also known as self-driving vehicles or connected and AVs (CAVs), can access Internet services if they connect to the roadside unit (RSU) through the vehicle-to-infrastructure (V2I) communication [3]. AVs also communicate with each other to form a vehicle-to-vehicle (V2V) network for the purpose of being aware of neighboring vehicles or being notified immediately when any emergent situation happens [4], [5].

V2V communication systems are composed of devices, installed in vehicles, that use dedicated short-range radio communication (DSRC) [6] supported by the IEEE 802.11p [7], IEEE 1609 family of standards [8], or 5G automotive association (5GAA) [9] cellular vehicle-to-everything (C-V2X) [10], a long-term evolution (LTE)-based radio access technologies (RAT) [11], [12], to exchange messages containing vehicle information (e.g., vehicle's speed, heading, and braking status). These RAT evolutions (the IEEE 802.11bd [13] for the DSRC and new radio (NR) V2X [14], [15] for C-V2X) are undergoing. V2V devices use this information from other vehicles and determine if a warning to the vehicle's driver is needed, which could prevent a vehicle crash. DSRC is two-way, wireless communications permitting secure and fast messaging needed for safety applications, where "short range" is approximately 300 m depending on the surrounding environment [16]. These communications occur in a 75-MHz band of the 5.9-GHz spectrum, which has been allocated by the FCC for use by the ITS vehicle safety and mobility applications [17].

The survey work [11] showed that C-V2X (ETSI LTE) and DSRC (IEEE 802.11p) technologies adopt the same spectrum between 5.855 and 5.925 GHz and each channel bandwidth is 10 MHz. Both technologies use the carrier-sense multiple access with collision avoidance (CSMA/CA) as the media access control (MAC) mechanism and the time-division duplex (TDD) method, while C-V2X uses synchronous transmission and DSRC uses asynchronous transmission [11]. Study works [18], [19] showed that C-V2X outperforms DSRC in almost all situations. In [17], the study revealed that C-V2X and DSRC are not compatible with each other and lead to coexistence problems.

When AVs which have ongoing V2V communication come within the coverage of an RSU, say A_f as shown in Fig. 1, the AVs suffer interference from the infrastructure-to-vehicle (I2V) links (i.e., the downlink of V2I communication) if the RSU has at least one authorized infrastructure users (IUs) in service. In this case, the communication between IUs and the RSU will be interfered by V2V communication, and *vice versa* [20]. According to the regulation of 5GAA [21] and Third Generation Partnership Project (3GPP) [22], the vehicle will perform broadcast report to the RSUs when entering or

Manuscript received July 14, 2019; revised October 30, 2019; accepted January 9, 2020. Date of publication January 17, 2020; date of current version May 12, 2020. This work was supported in part by the Ministry of Science and Technology, Taiwan, under Contract MOST 107-2221-E-182-072; and in part by the National Natural Science Foundation of China, China, under Grant 61671073. *(Corresponding author: Jenhui Chen.)*

Xiangming Wen and Zhaoming Lu are with the College of Information and Communication Engineering, Beijing University of Posts and Telecommunication, Beijing 100876, China (e-mail: xiangmw@bupt.edu.cn; lzy_0372@163.com).

Jenhui Chen is with the Department of Computer Science and Information Engineering, College of Engineering, Chang Gung University, Kweishan, Taoyuan 33302, Taiwan and also with the Center for Artificial Intelligence in Medicine, Chang Gung Memorial Hospital, Kweishan, Taoyuan 33302, Taiwan (e-mail: jhchen@mail.cgu.edu.tw).

Zhiqun Hu is with the College of Computer and Information Engineering, Hubei University, Wuhan 430062, China (e-mail: huzhiqun@bupt.edu.cn).

Digital Object Identifier 10.1109/JIOT.2020.2967647

2327-4662 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

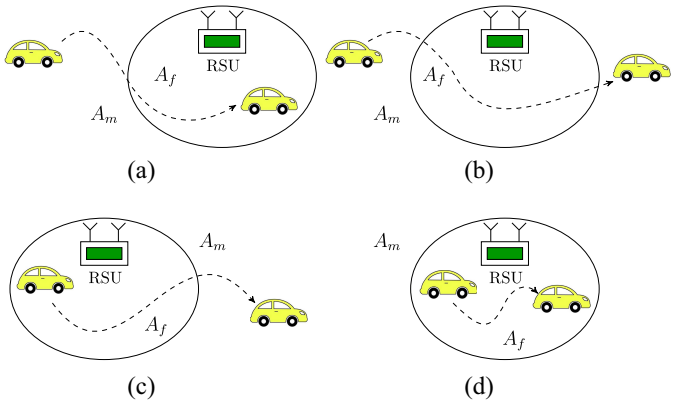


Fig. 1. Four cases of overlapping time t_o between the AV's ongoing V2V communication and RSU's channel access. (a) Case I. (b) Case II. (c) Case III. (d) Case IV.

leaving the coverage of an RSU. Then, the RSU will suspend access services to its IUs.

Let A_m denote the blind zone with A_f in the road. Suppose that RSUs are separated far enough so that the AV's journey along the road will visit A_m and A_f alternately (e.g., the journey could be $A_m \rightarrow A_{f_i} \rightarrow A_m \rightarrow A_{f_{i+1}} \rightarrow \cdots$). Fig. 1 illustrates four possible interference cases where the RSU will interfere with the V2V communication. It shows that the V2V communication initiates at A_m and ends at A_f in case I or the V2V communication ends at A_m in case II. Meanwhile, cases III and IV show that the V2V communication initiates at A_f and ends at A_m (case III) or at A_f (case IV). Here, we address the "co-channel" problem in scenarios that AVs and RSUs use: 1) the same protocol (e.g., C-V2X or 802.11p); 2) different versions of protocol (e.g., 802.11bd and 802.11p); or 3) a hybrid of them (e.g., 802.11p/bd and C-V2X).

Many interference management mechanisms had been broadly studied, such as network coding for mitigating the interference among V2V and V2I networks [23], decentralized resource allocation mechanism for the V2V communications based on deep reinforcement learning [24], [25], interference mitigating by successive interference cancellation scheme [26], self-interference cancellation scheme in a full-duplex network [27], interference reducing by minimizing the transmit power schemes [28]–[30], jointly optimizing radio resource power allocation [31], [32], and dividing high interfering V2V links into different clusters based on a graph algorithm [33]. Radio resource allocation strategies were proposed for vehicle networks to minimize the effect of interference [34], [35]. Several spectrum allocation mechanisms are used for spectrum splitting, such as preplanned frequency assignment approaches [36], the interference-aware graph approach [37], coalitional gaming approaches [38]–[40], matching gaming approaches [41], [42], and the cooperative spectrum allocation mechanism [43], [44]. These studies concluded that the most difficulty in dealing with the interference of vehicle networks is pertained to the mobility of vehicles.

The problem of dealing interference in the 2-tier architecture of vehicular networks (i.e., between V2V and V2I communications) has two difficulties. First, V2V links may meet V2I links frequently (e.g., in a hotspot area) or meet V2I scarcely (e.g., in a suburban or private area). This factor leads to the need for a dynamic setting of the RSUs depending on the frequency of bandwidth usage by AVs neighboring to the RSU.[1] Second, the RSUs cannot use the carrier-sense scheme or listen to the resource allocation information issued from the V2Vs to avoid collision with nearby V2Is in the downlink direction at the same time. The two main causes of difficulties prevent efficient interference avoidance of RSUs. Moreover, if the bandwidth utilization of these RSUs is put into account, these difficulties would be more complicated.

In this article, we study a *p*-opportunity frame-based channel access scheme (*p*-OCAS) which has similar concept of the well known *p*-persistent CSMA mechanism [45] but quite different from the *p*-persistent CSMA mechanism for tackling the problem of V2V communication interfered by I2V communication in the 2-tier vehicular networks. The main idea behind the *p*-OCAS is that it enables the RSU to access the channel at the commencement of a frame with probability p for a fixed channel access period t_2, or to defer it is access by a fixed time period t_1 with probability $1-p$, where $0 < p \leq 1$, as shown in Fig. 2(a). When the RSU successfully access the channel for t_2, the RSU will increase the value of p. Otherwise, at least one AV will experience a signal interference from the I2V communication, which will prompt the AV to send an *interference signal* to the RSU in the uplink frame as shown in Fig. 2(b). In this case, the RSU will decrease the value of p and delay duration of time $E[t_o]$ for keeping silent on downlink.

In the design mechanism, AVs are assigned a higher priority of channel access than RSUs because we assume that: 1) V2V communication conveys critical messages for car safety [21], [46], [47] and 2) the downlink bandwidth of RSU for IUs is sufficient [48]. This implies that the suspended time period of channel access can be compensated later by the RSU via bandwidth scheduling. The detailed operations of *p*-OCAS will be presented in Section II. The main contributions of this article are as follows.

1) We address the co-channel problem of I2V and V2V communications in the downlink of RSU because of the CSMA/CA mechanism and the hidden node problem in the 2-tier vehicular networks.
2) We propose a *p*-OCAS, which is a MAC layer solution to tackle the problem of AVs interfered by RSU. *p*-OCAS is capable of dynamically adjusting the downlink channel access probability according to the co-channel V2V communication neighboring to the RSU.
3) We present a theoretical analysis regarding the interference probability of the V2V and V2I communication, and study how the multiple parameters affect the throughput and the average MAC delay of RSU.

II. *p*-Opportunity Channel Access Scheme

Let $S_0, S_1, \ldots, S_n$ represent n states of the RSU and p_n be the channel access probability of the state S_n. Since the channel access attempt relies on the probability p_n, the RSU may delay several radio frames before it accesses the channel.

[1] Suppose the probability that AVs and the RSU use bandwidth at the same time is proportional to the number of AVs surrounding the RSU.

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

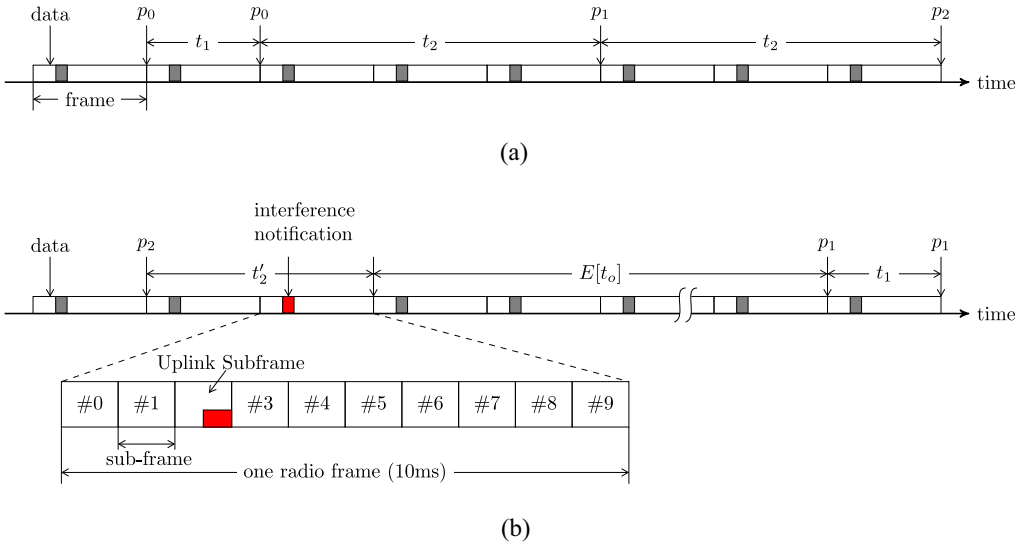


Fig. 2. Example of p-opportunity channel access scheme. The radio frame structure follows the LTE-A specification. (a) Successful channel access example. (b) Illustration of frame access operation as interference between V2V and V2I.

If the RSU has packets to send, it will send the packet with a probability p_n on the channel at the commencement of a frame for a duration t_2 (e.g., several frames) or defer a duration t_1 (e.g., a frame) with probability $1 - p_n$. Fig. 2(a) shows an example that the first attempt of channel access does not succeed with probability p_0 but instead succeeds at the second attempt with probability p_0. Let X_n be the random variable that the number of attempts the RSU tries with p_n to access the channel. Obviously, the number of attempts X_n that the first occurrence of success requires k number of independent trials each with a success probability p_n, follows the geometric distribution. The probability that the RSU gets the right to access the channel at the kth attempt is

$$P[X_n = k] = (1 - p_n)^{k-1} p_n, \quad k = 1, 2, 3, \ldots \tag{1}$$

where $0 < p_n \leq 1$. If the access of the channel with p_n is successful [i.e., no interference occurs in this trial as shown in the first t_2 period of Fig. 2(a)], it uses probability p_{n+1} to access the channel [in this example, $p_{n+1} = p_1$]. If the trial of S_n fails, the probability is set as the probability of S_{n-1}. The probability function $f_p(i)$ of state S_i is given by

$$f_p(i) = \min\left(1, 2^i p_0\right) \tag{2}$$

where p_0 is the initial probability (i.e., the probability of S_0). The interference notification (interfering with AVs) is informed by the interfered AVs if any. Fig. 2(b) shows the interference notification (see the red-colored square) sent from the interfered AV in the uplink subframe during the period t_2 (denoted as t_2' to indicate collisions occurring during t_2).

The design of an interference notification slot is described as follows. Each radio frame (frame for short) follows the frame structure of LTE-A specification [18], [49] as shown in Fig. 2. The frame structure could be type 1 (FDD) or type 2 (TDD). The uplink physical random access channel (PRACH) is used to let the AV send the interference notification to the RSU. The network can also notify the AV of a reserved preamble to the neighboring RSU as specified in [49] when the AV performs proximity report. p-OCAS sends the assigned preamble sequence number to indicate to the neighboring RSU about the interference once the AV is interfered.

Fig. 2 demonstrates channel access examples of p-OCAS. Fig. 2(a) shows the example of successful channel access. The RSU uses two attempts with p_0 to obtain a channel access trial (i.e., a channel access opportunity) for a period of t_2. Since no AVs stay in the radio coverage of the RSU and happens to have V2V communication sessions (V2V sessions for short) during t_2, the RSU successfully accesses the channel for t_2 and will access channel with p_1 at the commencement of next frame after the end of t_2. Fig. 2(b) shows the interfered example. The RSU obtains the channel access right with p_2 but gets an interference notification from an AV; the RSU abandons the access of channel immediately and delays for duration $E[t_o]$, and then sets its p be $p_1 = f_p(2 - 1)$. The number of states $S_0, S_1, \ldots, S_n$ is equal to $n + 1$. The value of n can be determined by initial probability p_0 and is given by

$$n = \left\lceil \log_2 p_0^{-1} \right\rceil. \tag{3}$$

The value of $E[t_o]$ is related to the expected V2V session duration and the V2V session arrival rate to AV and the conditional probability of AV's appearance within the RSU's coverage A_f. The notations used for p-OCAS are shown in Table I and derivation of $E[t_o]$ is illustrated in Section III.

III. Analytical Model

The system is modeled as a vehicular network where the respective V2V and V2I communication links coexist. Suppose that an AV arrives in an RSU's coverage A_f with t_f in accordance with a gamma distribution with a rate λ_f [51]. Upon arrival to A_f (i.e., the AV can receive signals from the RSU), the AV will immediately interfere with the RSU for a period of time if the AV happens to have an ongoing V2V session while the RSU serves some IUs. The period of time is the

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

overlapping time t_o that the AV stays in A_f with an ongoing V2V session while the RSU is accessing the same channel. Let the residence time of an AV in A_m be t_m with a gamma distribution with a rate λ_m. Also, consider the time interval t_h between two consecutive V2V sessions of the AV. Thus, for $i \geq 1$, let $t_{f,i}$ be the ith duration that AV visits A_f after the previous V2V session request arrival, and let $t_{m,i}$ be the ith duration that AV visits A_m after the previous V2V session request arrival. Fig. 3 illustrates four possible cases of t_o as shown in Fig. 1.

Fig. 3(a) shows case I of Fig. 1 where a V2V session arrives to/from an AV when the AV is within the areas of either A_m or A_f and completes the V2V session in A_f finally. Fig. 3(b) shows case II of Fig. 1 where the V2V session arrives to/from an AV begins and completes its session in either A_m or A_f and t_h spans at least one t_f. Furthermore, Fig. 3(c) shows case III of Fig. 1, where the V2V session arriving to/from an AV begins in A_f and ends in another A_m or A_f. Finally, Fig. 3(d) shows case IV that t_h starts and ends at the same A_f.

In the analytical model, we make the following assumptions.

1) The incoming V2V session to/from an AV are a Poisson process with a mean rate λ [i.e., the interarrival time t is exponentially distributed with the density function $f(t) = \lambda e^{-\lambda t}$].
2) The V2V session duration is exponentially distributed with mean $1/\mu$. The density function is $f_h(t) = \mu e^{-\mu t}$.
3) The overlapped time random variable t_o is exponentially distributed with mean $1/\lambda_o$ and the density function $f_o(t) = \lambda_o e^{-\lambda_o t}$.
4) The probability that RSU will access the channel is ρ.
5) The probability that AV will access the channel is $\mathcal{P}_m = E[t_h]/E[t_c] = \lambda/u$.
6) $t_{m,i}$ and $t_{f,i}$ random variables are independent and identically distributed (i.i.d.) with the density function $f_m(t)$ and $f_f(t)$, the mean $1/\lambda_m$ and $1/\lambda_f$, and the Laplace transform $f_m^*(s)$ and $f_f^*(s)$, respectively.
7) According to the renewal process, the probability of entering A_f and A_m, denoted as p_f and p_m, is

$$p_f = \frac{E[t_{f,i}]}{E[t_{m,i}] + E[t_{f,i}]} = \frac{\lambda_m}{\lambda_f + \lambda_m}$$

and

$$p_m = \frac{E[t_{m,i}]}{E[t_{m,i}] + E[t_{f,i}]} = \frac{\lambda_f}{\lambda_f + \lambda_m}$$

respectively.

The studied output metrics in this article are:

1) p_c*:* the probability of V2Vs being interfered by RSU;
2) *S:* the throughput of RSU;
3) *E(T_M):* the average MAC delay of RSU.

A. Derivation of p_c

Let N_f be the number of RSU areas that AV visits during t_h. We consider four cases to derive $\Pr[N_f = k]$ for $k \geq 1$.

Case I: The previous V2V session arrives when the AV is in A_m and the V2V session ends when the AV is in A_f as shown in Fig. 3(a).

Case II: The previous V2V session arrival to the AV begins and completes in A_m as shown in Fig. 3(b).

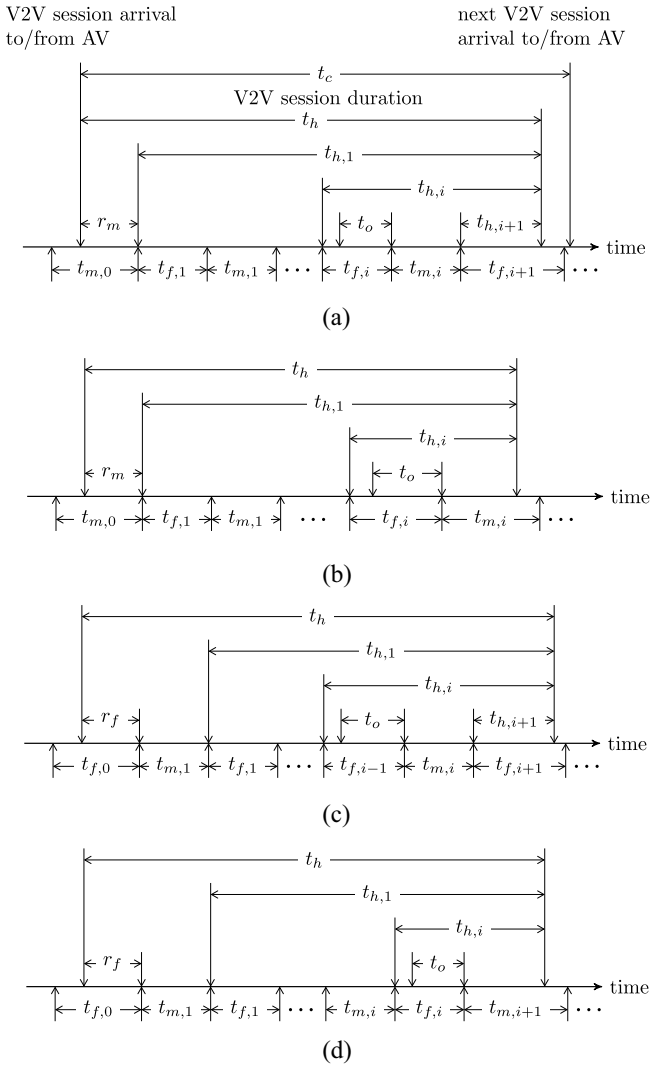


Fig. 3. Time diagram of four possible cases of overlapping time t_o between the AV's ongoing V2V session and RSU's channel access. The illustration of session time intervals of (a) case I; (b) case II; (c) case III; and (d) case IV.

Case III: The previous V2V session arrives when the AV is in A_f and the V2V session ends when the AV is in A_f as shown in Fig. 3(c).

Case IV: The previous V2V session arrives when the AV is in A_f and the V2V session ends when the AV is in A_m as shown in Fig. 3(d).

Let $N_{\mathrm{I},f}$, $N_{\mathrm{II},f}$, $N_{\mathrm{III},f}$, and $N_{\mathrm{IV},f}$ be the number of N_f conditioning on cases I–IV, respectively. Then, $\Pr[N_f = k]$ is given by

$$\begin{aligned}\Pr[N_f = k] = {} & \Pr[N_{\mathrm{I},f} = k] \Pr[\text{case I}] \\ & + \Pr[N_{\mathrm{II},f} = k] \Pr[\text{case II}] \\ & + \Pr[N_{\mathrm{III},f} = k] \Pr[\text{case III}] \\ & + \Pr[N_{\mathrm{IV},f} = k] \Pr[\text{case IV}]. \end{aligned} \tag{4}$$

According to the above assumptions, we have

$$\Pr[\text{case I}] = p_m p_f = \frac{\lambda_f \lambda_m}{(\lambda_f + \lambda_m)^2} \tag{5}$$

$$\Pr[\text{case II}] = p_m p_m = \frac{\lambda_f^2}{(\lambda_f + \lambda_m)^2} \tag{6}$$

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NOTATIONS USED FOR p-OCAS AND ANALYSIS

Notation	Description
N_f	The number of RSU areas that AV visits during t_h
A_f	The coverage of RSU
A_m	The coverage which is not covered by RSU
λ_f	The AV arrival rate in the RSU's coverage
λ_m	The AV arrival rate in blind zone
λ_o	The AV arrival rate when the V2V session arriving
λ	The mean rate for V2V sessions of AVs
μ	The mean V2V session duration
ρ	The probability that RSU access the channel
p_n	The channel access attempt probability of RSU at state S_n
p_0	The initial channel access attempt probability of RSU at state S_0
$\mathcal{P}_m$	The probability that the AV will access the channel
p_f	The probability of entering A_f
p_m	The probability of entering A_m
$t_{m,i}$	The i-th residence time that AV visits A_m
$t_{f,i}$	The i-th residence time that AV visits A_f
t_1	The backoff time for RSU
t_2	The channel access period for RSU
t_2'	The collision duration between RSU and AVs
t_h	The V2V session holding duration of AV
t_c	The time interval between 2 consecutive V2V sessions
t_o	The overlapping time that the AV stays in A_f with an ongoing V2V session while the RSU is accessing the same channel
T_F	The duration of a LTE-A frame
D_i	The MAC delay of p-OCAS at state S_i
$E[t_o]$	The average deferred time duration
r_m	The time period that the AV visits the A_m with V2V session arrival
r_f	The time period that the AV visits the A_f with V2V session arrival
N_f	The number of RSU areas that AV visits during the V2V session
ρ	The expected probability that the RSU is allowed to access the channel
S	The throughput of RSU
p_c	The probability of V2V being interfered by RSU
$E(T_M)$	The average MAC access delay of RSU

$$\Pr[\text{case III}] = p_f p_f = \frac{\lambda_m^2}{\left(\lambda_f + \lambda_m\right)^2} \tag{7}$$

and

$$\Pr[\text{case IV}] = p_f p_m = \frac{\lambda_m \lambda_f}{\left(\lambda_f + \lambda_m\right)^2}. \tag{8}$$

Let r_m be the time period that the AV visits A_m for the first time and the V2V session arrives at A_m. Under the conditions of cases I and II, let $t_{M,k} = r_m + \sum_{i=1}^{k}(t_{f,i} + t_{m,i})$ and $t_{F,k} = r_m + t_{f,1} + \sum_{i=1}^{k}(t_{m,i} + t_{f,i+1})$. By convention, $t_{M,0} = r_m$ and $t_{F,0} = r_m + t_{f,1}$. $t_{M,k}$ and $t_{F,k}$ are random variables with the general density function $f_{M,k}(t)$ and $f_{F,k}(t)$ and the Laplace transform $f_{M,k}^*(s)$ and $f_{F,k}^*(s)$, respectively. From the convolution rule of the Laplace transform, we get

$$f_{M,k}^*(s) = r_m^*(s)\left[f_f^*(s) f_m^*(s)\right]^k \tag{9}$$

and

$$f_{F,k}^*(s) = r_m^*(s) f_f^*(s)\left[f_f^*(s) f_m^*(s)\right]^k \tag{10}$$

where $r_m^*(s)$ is the Laplace transform of the random variable r_m with density function $r_m(t)$. According to [51] and [52], the Laplace transform of $r_m(t)$ is obtained by

$$r_m^*(s) = \left(\frac{\lambda_m}{s}\right)\left[1 - f_m^*(s)\right]. \tag{11}$$

In case I, the previous V2V session arrives during r_m, and the next V2V session arrives during $t_{f,j}$ where $j \geq 1$. In this case, $t_{M,j-1} < t_h < t_{F,j-1}$. Then, we have

$$\begin{aligned}\Pr\left[N_{\mathrm{I},f} = j\right] &= \Pr\left[t_{M,j-1} < t_h < t_{F,j-1}\right] \\ &= \Pr\left[t_h > t_{M,j-1}\right] - Pr\left[t_h > t_{F,j-1}\right] \\ &= \int_{t_{M,j-1}=0}^{\infty}\int_{t_h=t_{M,j-1}}^{\infty} f_{M,j-1}\left(t_{M,j-1}\right) \\ &\quad \times \mu e^{-\mu t_h} dt_{M,j-1} dt_{\mathrm{h}} \\ &\quad - \int_{t_{F,j-1}=0}^{\infty}\int_{t_h=t_{F,j-1}}^{\infty} f_{F,j-1}\left(t_{F,j-1}\right) \\ &\quad \times \mu e^{-\mu t_h} dt_{F,j-1} dt_h \\ &= f_{M,j-1}^*(\mu) - f_{F,j-1}^*(\mu). \end{aligned} \tag{12}$$

After arranging, we obtain

$$\begin{aligned}\Pr\left[N_{\mathrm{I},f} = j\right] = \left(\frac{\lambda_m}{\mu}\right)\left[1 - f_m^*(\mu)\right]\left[1 - f_f^*(\mu)\right] \\ \times \left[f_f^*(\mu) f_m^*(\mu)\right]^{j-1}, \quad j \geq 1. \end{aligned} \tag{13}$$

In case II, the previous V2V session arrives during r_m, and the next V2V session arrives during $t_{m,j}$ where $j \geq 0$. In this case, $t_{F,j-1} < t_h < t_{M,j}$, $j \geq 1$. Then, we have

$$\begin{aligned}\Pr\left[N_{\mathrm{II},f} = j\right] &= \Pr\left[t_{F,j-1} < t_h < t_{M,j}\right] \\ &= \Pr\left[t_h > t_{F,j-1}\right] - \Pr\left[t_h > t_{M,j}\right] \\ &= \int_{t_{F,j}=0}^{\infty}\int_{t_h=t_{F,j}}^{\infty} f_{F,j-1}\left(t_{F,j-1}\right)\mu e^{-\mu t_h} dt_{F,j-1} dt_h \\ &\quad - \int_{t_{M,j}=0}^{\infty}\int_{t_h=t_{M,j}}^{\infty} f_{M,j}\left(t_{M,j}\right)\mu e^{-\mu t_h} dt_{M,j} dt_h \\ &= f_{F,j-1}^*(\mu) - f_{M,j}^*(\mu). \end{aligned} \tag{14}$$

When $j = 0$, we have

$$\begin{aligned}\Pr\left[N_{\mathrm{II},f} = 0\right] &= \Pr[t_h < \mathrm{r}_m] \\ &= \int_{t=0}^{\infty}\int_{0}^{t_h=t} r_m(t)\mu e^{-\mu t_h} dt dt_h = 1 - r_m^*(u). \end{aligned} \tag{15}$$

After arranging, we obtain

$$\begin{aligned}&\Pr\left[N_{\mathrm{II},f} = j\right] \\ &= \begin{cases} \left(\frac{\lambda_m}{\mu}\right) f_f^*(\mu)\left[1 - f_m^*(\mu)\right]^2\left[f_f^*(\mu) f_m^*(\mu)\right]^{j-1}, & j \geq 1 \\ 1 - r_m^*(u), & j = 0. \end{cases} \end{aligned} \tag{16}$$

Let r_f be the time period that the AV visits A_f for the first time and the V2V session arrives at A_f. For cases III and IV, let $t_{M1,k} = r_f + t_{m,1} + \sum_{i=1}^{k}(t_{f,i} + t_{m,i+1})$ and $t_{F1,k} = r_f + \sum_{i=1}^{k}(t_{m,i} + t_{f,i})$. By convention, $t_{M1,0} = r_f + t_{m,1}$ and $t_{F1,0} = r_f$. $t_{M1,k}$ and $t_{F1,k}$ are random variables with the general density function $f_{M1,k}(t)$ and $f_{F1,k}(t)$

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

and the Laplace transform $f^*_{M1,k}(s)$ and $f^*_{F1,k}(s)$, respectively. From the convolution rule of the Laplace transform, we get

$$f^*_{M1,k}(s) = r^*_f(s) f^*_m(s) \Big[f^*_f(s) f^*_m(s) \Big]^k \tag{17}$$

and

$$f^*_{F1,k}(s) = r^*_f(s) \Big[f^*_f(s) f^*_m(s) \Big]^k \tag{18}$$

where $r^*_f(s)$ is the Laplace transform of the random variable r_f with density function $r_f(t)$. According to [51] and [52], the Laplace transform of $r_f(t)$ is obtained by

$$r^*_f(s) = \left(\frac{\lambda_f}{s} \right) \Big[1 - f^*_f(s) \Big]. \tag{19}$$

Considering cases III and IV, the derivation for $\Pr[N_{\mathrm{III},f} = k]$ and $\Pr[N_{\mathrm{IV},f} = k]$ is similar to that for cases I and II. Then, we have

$$\Pr\big[N_{\mathrm{III},f} = j\big] = \begin{cases} \left(\frac{\lambda_f}{\mu} \right) f^*_m(\mu) \Big[1 - f^*_f(\mu) \Big]^2 \Big[f^*_f(\mu) f^*_m(\mu) \Big]^{j-2}, & j >= 2 \\ 1 - r^*_f(u), & j = 1 \end{cases} \tag{20}$$

and

$$\Pr\big[N_{\mathrm{IV},f} = j\big] = \left(\frac{\lambda_f}{\mu} \right) \Big[1 - f^*_f(\mu) \Big] \big[1 - f^*_m(\mu) \big] \times \Big[f^*_f(\mu) f^*_m(\mu) \Big]^{j-1}, \quad j \geq 1. \tag{21}$$

Now, we consider the case that the RSU accesses the channel will contribute in the interference to AV. Let α be the probability that RSU accesses the channel during $t_{f,i}$ under the condition that the V2V session does not end during $t_{f,i}$. Then, we get

$$\begin{aligned} \alpha &= \Pr\big[\text{RSU accesses the channel, } t_o < t_{f,i} | t_{h,i} > t_{f,i}\big] \\ &= \frac{\Pr\big[\text{RSU accesses the channel, } t_o < t_{f,i} < t_{h,i}\big]}{\Pr\big[t_{h,i} > t_{f,i}\big]} \\ &= \frac{\rho \Big[f^*_f(u) - f^*_f(u + \lambda_o) \Big]}{f^*_f(u)}. \end{aligned} \tag{22}$$

Let β be the probability that RSU accesses the channel under the condition that AV completes the V2V session during $t_{f,i}$, and we have

$$\begin{aligned} \beta &= \Pr\big[\text{RSU accesses the channel, } t_o < t_{h,i} | t_{h,i} < t_{f,i}\big] \\ &= \frac{\Pr\big[\text{RSU accesses the channel, } t_o < t_{h,i} < t_{f,i}\big]}{\Pr\big[t_{h,i} < t_{f,i}\big]} \\ &= \frac{\Pr[\text{RSU accesses the channel}] \Pr\big[t_0 < t_{h,i} < t_{f,i}\big]}{\Pr\big[t_{h,i} < t_{f,i}\big]} \\ &= \frac{\rho \Big[\left(\frac{\lambda_o}{u + \lambda_o} \right) + \left(\frac{u}{u + \lambda_o} \right) f^*_f(u + \lambda_o) - f^*_f(u) \Big]}{1 - f^*_f(u)}. \end{aligned} \tag{23}$$

Let φ be the probability that RSU accesses the channel on the condition that AV starts the V2V session during r_f, and we have

$$\begin{aligned} \varphi &= \Pr\big[\text{RSU accesses the channel, } t_o < r_f | t_h \geq r_f\big] \\ &= \frac{\Pr\big[\text{RSU accesses the channel, } t_o < r_f < t_h\big]}{\Pr\big[t_h \geq r_f\big]} \\ &= \frac{\rho \Big[r^*_f(u) - r^*_f(u + \lambda_o) \Big]}{r^*_f(u)}. \end{aligned} \tag{24}$$

Without the loss of generality, we can express the interference probability of AV p^j_c that at least one of the RSU accesses the channel under the case that the number of crossings A_f for the AV during t_h is j. Then, p^j_c is shown in (25), at the bottom of the page. Therefore, the average interference probability for AV, p_c is given by

$$p_c = \sum_{j=0}^{\infty} p^j_c \mathcal{P}_m = \sum_{j=0}^{\infty} p^j_c \left(\frac{\lambda}{\mu} \right). \tag{26}$$

In this article, the gamma distribution for the residence time t_f and t_m is selected because it can approximate many types of distributions. It has a closed-form expression for its Laplace transform, and it has been widely used in previous works [51], [52]. Suppose that the gamma density functions $f_f(t)$ and $f_m(t)$ are with the mean $1/\lambda_f$ and $1/\lambda_m$, the shape parameters γ_f and γ_m, and the variances $v_f = 1/(\gamma_f \lambda_f^2)$ and $v_m = 1/(\gamma_m \lambda_m^2)$, respectively. Then, the Laplace transform is

$$f^*_f(s) = \left(\frac{\gamma_f \lambda_f}{s + \gamma_f \lambda_f} \right)^{\gamma_f} \text{and} \quad f^*_m(s) = \left(\frac{\gamma_m \lambda_m}{\gamma_m \lambda_m + s} \right)^{\gamma_m}. \tag{27}$$

B. Derivation of E[t_o]

In this section, we derive the expected overlapping time $E[t_o]$ that an AV has an ongoing V2V session and stays in A_f concurrently. Suppose t_f is i.i.d. random variables, the probability that $t_o \leq t_f$ is obtained by

$$\begin{aligned} P\big[t_o \leq t_f\big] &= \int_{t_f=0}^{\infty} \int_{t_o=0}^{t_f} \lambda_o e^{-\lambda_o t_o} f_f\big(t_f\big) dt_o dt_f \\ &= \int_{t_f=0}^{\infty} \big(1 - e^{-\lambda_o t_f}\big) f_f\big(t_f\big) dt_f \\ &= \int_{t_f=0}^{\infty} f_f\big(t_f\big) dt_f - \int_{t_f=0}^{\infty} e^{-\lambda_o t_f} f_f\big(t_f\big) dt_f \\ &= 1 - f^*_f(\lambda_o) \end{aligned} \tag{28}$$

$$p^j_c = \begin{cases} \big(1 - (1-\alpha)^{j-1}(1-\beta)\big) \Pr\big[N_{\mathrm{I},f} = j\big] \Pr[\text{case I}] + \big(1 - (1-\alpha)^j\big) \Pr\big[N_{\mathrm{II},f} = j\big] \Pr[\text{case II}] \\ \quad + \big(1 - (1-\alpha)^{j-2}(1-\varphi)(1-\beta)\big) \Pr\big[N_{\mathrm{III},f} = j\big] \Pr[\text{case III}] \\ \quad + \big(1 - (1-\alpha)^{j-1}(1-\varphi)\big) \Pr\big[N_{\mathrm{IV},f} = j\big] \Pr[\text{case IV}] & j \geq 2 \\ \beta \Pr\big[N_{\mathrm{I},f} = 1\big] \Pr[\text{case I}] + \alpha \Pr\big[N_{\mathrm{II},f} = 1\big] \Pr[\text{case II}] + \varphi \Pr\big[N_{\mathrm{IV},f} = 1\big] \Pr[\text{case IV}] & j = 1 \end{cases} \tag{25}$$

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

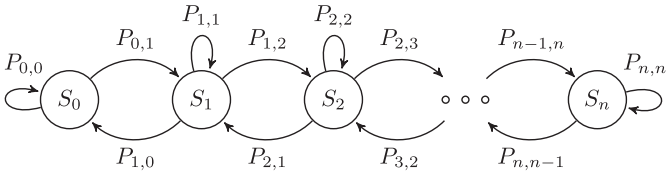


Fig. 4. Semi-Markov chain model of a CF with the p-OCAS.

and the probability that $t_o > t_f$ is

$$\begin{aligned} P\big[t_o > t_f\big] &= \int_{t_f=0}^{\infty}\int_{t_o=t_f}^{\infty} \lambda_o e^{-\lambda_o t_o} f_f\big(t_f\big) dt_o dt_f \\ &= \int_{t_f=0}^{\infty} e^{-\lambda_o t_f} f_f\big(t_f\big) dt_f \\ &= f_f^*(\lambda_o). \end{aligned} \tag{29}$$

According to (28) and (29), the expected value of t_o is obtained by

$$\begin{aligned} E[t_o] &= t_o P\big[t_o \le t_f\big] + t_f P\big[t_o > t_f\big] \\ &= \int_{t_f=0}^{\infty}\int_{t_o=0}^{t_f} t_o \lambda_o e^{-\lambda_o t_o} f_f\big(t_f\big) dt_o dt_f \\ &\quad + \int_{t_f=0}^{\infty}\int_{t_o=t_f}^{\infty} t_f \lambda_o e^{-\lambda_o t_o} f_f\big(t_f\big) dt_o dt_f \\ &= \int_{t_f=0}^{\infty} -t_o e^{-\lambda_o t_o} - \frac{1}{\lambda_o} e^{-\lambda_o t_o}|_{t_o=0}^{t_f} \cdot f_f\big(t_f\big) dt_f \\ &\quad + \int_{t_f=0}^{\infty} -e^{-\lambda_o t_o}|_{t_o=t_f}^{\infty} \cdot t_f f_f\big(t_f\big) dt_f \\ &= \frac{1 - f_f^*(\lambda_o)}{\lambda_o}. \end{aligned} \tag{30}$$

Integrating (27) into (30), we have

$$E[t_o] = \frac{1 - \left(\frac{\gamma_f \lambda_f}{\lambda_o + \gamma_f \lambda_f}\right)^{\gamma_f}}{\lambda_o}. \tag{31}$$

C. Semi-Markov Chain Model

To analyze the RSU channel access process, a semi-Markov chain model is adopted. Fig. 4 shows the semi-Markov chain model of p-OCAS in which $n = \lceil \log_2 p_0^{-1} \rceil$. Let $P_{i,j}$ be the transition probability of the RSU from S_i to S_j and π_i denote the long-term fraction of time that the RSU stays in S_i. The state transition probabilities of the semi-Markov chain for $1 \le i \le n-1$ is written in component forms as

$$P_{i,i} = 1 - f_p(i) \tag{32}$$

$$P_{i,i+1} = f_p(i) e^{-\lambda_o t_2} \tag{33}$$

and

$$P_{i,i-1} = f_p(i)\big(1 - e^{-\lambda_o t_2}\big). \tag{34}$$

The state transition probabilities of the semi-Markov chain for $i = 0$ are

$$P_{0,0} = 1 - p_0 e^{-\lambda_o t_2} \tag{35}$$

and

$$P_{0,1} = p_0 e^{-\lambda_o t_2}. \tag{36}$$

The state transition probabilities of the semi-Markov chain for $i = n$ are expressed as

$$P_{n,n} = 1 - f_p(n)\big(1 - e^{-\lambda_o t_2}\big) \tag{37}$$

and

$$P_{n,n-1} = f_p(n)\big(1 - e^{-\lambda_o t_2}\big). \tag{38}$$

Using the equilibrium of the steady state, we have

$$\pi_0 P_{0,1} + \pi_0 P_{0,0} = \pi_1 P_{1,0} + \pi_0 P_{0,0} \tag{39}$$

$$\pi_n P_{n,n-1} + \pi_n P_{n,n} = \pi_{n-1} P_{n-1,n} + \pi_n P_{n,n} \tag{40}$$

and

$$\begin{aligned} &\pi_{i-1} P_{i-1,i} + \pi_i P_{i,i} + \pi_{i+1} P_{i+1,i} \\ &\quad = \pi_i P_{i,i-1} + \pi_i P_{i,i} + \pi_i P_{i,i+1} \quad (0 < i < n). \end{aligned} \tag{41}$$

According to the axiom of probability theory, $\pi_0 + \pi_1 + \cdots + \pi_n = 1$, it follows that:

$$\pi_0 = \left(1 + \sum_{j=1}^{n} \prod_{i=1}^{j} \frac{P_{i-1,i}}{P_{i,i-1}}\right)^{-1}. \tag{42}$$

Based on (39)–(42), the stationary state probability π_i is obtained by

$$\pi_i = \frac{P_{i-1,i}}{P_{i,i-1}} \pi_{i-1} = \prod_{j=1}^{i} \frac{P_{j-1,j}}{P_{j,j-1}} \pi_0. \tag{43}$$

Let $X_t(n)$ be the random variable of the duration the RSU stays in the state S_n. The expected value of $X_t(n)$ for each S_n, $n = 0, 1, \ldots, \lceil \log_2 p_0^{-1} \rceil$ is obtained by

$$\begin{aligned} E[X_t(n)] &= t_1\big(1 - f_p(n)\big) + t_2 f_p(n) e^{-\lambda_o t_2} \\ &\quad + \big(E\big[t_2'\big] + E[t_o]\big) f_p(n)\big(1 - e^{-\lambda_o t_2}\big) \end{aligned} \tag{44}$$

where

$$E\big[t_2'\big] = \frac{1}{t_2}(1 + 2 + \cdots + t_2) = \frac{t_2 + 1}{2} \tag{45}$$

is the mean value of t_2 (unit: frames). Based on (43) and (44), the expected operating time is obtained by

$$T_{\text{total}} = \sum_{i=0}^{n} \pi_i E[X_t(i)]. \tag{46}$$

$$\begin{aligned} S &= \frac{\sum_{i=0}^{n} \pi_i \big[t_2 f_p(n) e^{-\lambda_o t_2} + (1 + t_2) f_p(n)\big(1 - e^{-\lambda_o t_2}\big)/2\big]}{\sum_{i=0}^{n} \pi_i E[X_t(i)]} \\ &= \frac{\big[(t_2 - 1) e^{-\lambda_o t_2} + (1 + t_2)\big]}{2\sum_{i=0}^{n} \pi_i \Big[t_1 \frac{(1 - f_p(n))}{f_p(n)} + t_2 e^{-\lambda_o t_2} + \big(E\big[t_2'\big] + E[t_o]\big)\big(1 - e^{-\lambda_o t_2}\big)\Big]} \end{aligned} \tag{48}$$

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PARAMETERS FOR SIMULATION AND NUMERICAL RESULTS

Notation	Definition	Value
λ_f	AV arrival rate in a RSU's coverage	10, 20, 30
λ_o	AV arrival rate when V2V sessions arriving	$10^{-2} \sim 10^{0}\lambda_f$
$1/u$	Expected conversation period time	$1/5$ s
λ_m	AV arrival rate in blind zone	5
t_1	backoff time	0.01s
t_2	active	0.02s , 0.05s, 0.1s
p_0	initial probability to access channel	0.1, 0.2, 0.4

The expected probability that the RSU is allowed to access the channel, denoted as ρ, is obtained by

$$\rho = \sum_{i=0}^{\lceil \log_2 p_0^{-1} \rceil} \frac{\pi_i \big(f_p(i)t_2 + \big(E[t_2'] + E[t_o]\big) f_p(n)\big(1 - e^{-\lambda_o t_2}\big)\big)}{E[X_t(i)]}. \tag{47}$$

Since the expected duration of access trial in each S_n is

$$t_2 f_p(n) e^{-\lambda_o t_2} + (1 + t_2) f_p(n)\big(1 - e^{-\lambda_o t_2}\big)/2$$

the normalized throughput of RSU by using p-OCAS is obtained by (48), shown at the bottom of the previous page.

Let D_i denote the MAC delay of p-OCAS at state S_i. the MAC delay of p-OCAS at S_i is expressed as

$$\begin{aligned} D_i &= \frac{T_F}{2} + \sum_{k=1}^{\infty} f_p(i)\big(1 - f_p(i)\big)^{k-1}(k-1)t_1 \\ &= \frac{T_F}{2} + \frac{1 - f_p(i)}{f_p(i)} t_1 \end{aligned} \tag{49}$$

where T_F is the duration of a frame, e.g., $T_F = 10$ ms in the LTE-A system. According to (49), the average MAC delay is obtained by

$$E[T_M] = \sum_{i=1}^{n} \pi_i D_i. \tag{50}$$

IV. NUMERICAL RESULTS

In this section, a series of simulations was provided to validate the numerical results. The details of these simulations' design and flowchart are presented in the Appendix. Table II shows the parameters used in our simulation and numerical studies. First, we studied the performance of p-OCAS in the vehicular network, including V2V and V2I communication links. We focused on the evaluation of the effect of different parameters on the performance. The major output metrics are p_c, S, and $E[T_M]$, respectively. Second, the proposed p-OCAS for RSU is compared to both the p-random and p-persistent channel access schemes. The p-random channel access scheme illustrated that RSU chose a random number from [0, 1] as the probability to access the channel when RSU had a packet to transmit. On the other hand, the p-persistent channel access scheme presented the situation whereby the RSU accessed the channel with a fixed probability p.

First, we designed a simulation to validate our analysis model. The Appendix details how we implemented p-OCAS.

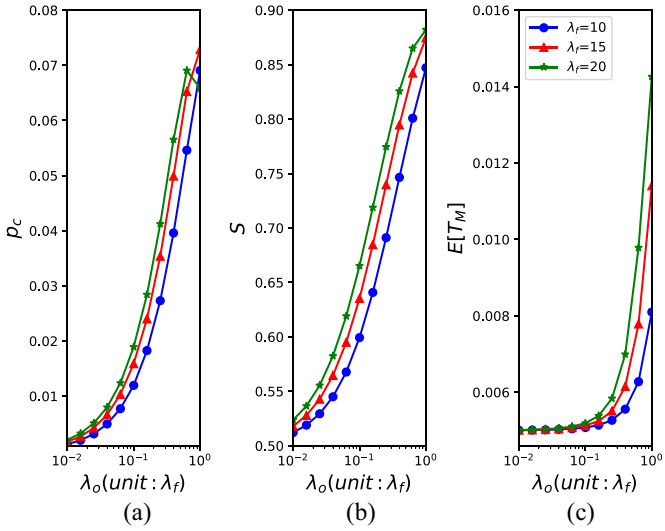


Fig. 5. Impact of λ_f on p_c, S, and $E[T_M]$ ($t_1 = 0.01$, $t_2 = 0.02$, $p_0 = 0.1$, $\gamma_m = 1/20$, and $\gamma_f = 1/20$). (a) Average interference probability of V2V. (b) Throughput of RSU. (c) Average MAC delay of RSU.

The simulation was implemented by the Python program. The flowchart of p-OCAS is given in Fig. 10. Table III shows the comparison between analysis and simulation results under $\lambda_f = 10$, 15, and 20, where $\lambda_o = 10^{-2}$, 10^{-1}, and 10^{0}, respectively. The observed measures were p_c, S, and $E[T_M]$. The initial channel access probability p_0 was set to 0.1. The backoff time and channel access duration for RSU were set to 0.01 and 0.02 s, respectively. A higher value of λ_f means higher arrival rate of AVs to the RSU's coverage. Table III shows that the results obtained from the analysis model are close to the simulation results in each performance metric.

Fig. 5 shows the impact of λ_f on p_c, S, and $E[T_M]$. As shown in Fig. 5, when $\lambda_f = 10$ (i.e., a lower arrival rate of AV to RSU's coverage), p_c increases as λ_o increases. The value of p_c is about 6.9% as shown in Fig. 5(a), when $\lambda_o = \lambda_f$. Furthermore, when the AVs arrive more quickly (e.g., $\lambda_f = 20$), p_c does not increases as λ_o increases. This is because p-OCAS lets RSUs terminate the access service to their subscribers for a designated period $E[t_o]$. A second incoming AV may arrive at an RSU during this period, and as a result, p_c will tend to decrease slightly while $\lambda_o = \lambda_f$. In Fig. 5(b), the value of S under $\lambda_f = 20$ is slightly higher than the value under $\lambda_f = 10$, because the duration that an AV stays in the coverage of the RSU is lower. However, S slowly increases when λ_o reaches up to a certain value. This is because an RSU should defer for long period $E[t_o]$ to avoid interference with nearby AVs. Fig. 5(c) shows that a higher value of λ_f results in a higher value of $E[T_M]$. Taking $\lambda_o = \lambda_f$ for example, the value of $E[T_M]$ is about 0.014 s under $\lambda_f = 20$ and 0.008 s under $\lambda_f = 10$. This is because higher λ_f results to the higher probability that an AV will come into the coverage of an RSU which has not attained the state S_n.

Fig. 6(a) shows that a higher initial channel access probability p_0 (e.g., $p_0 = 0.4$) will result in higher p_c. Obviously, it is easy to observe that a higher p_0 will cause interference with AVs easily. Although p-OCAS can let the RSU turn its state from S_i to S_{i-1} (i.e., the channel access probability becomes

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISONS BETWEEN SIMULATION RESULTS AND NUMERIC RESULTS AMONG INTERFERED PROBABILITY, THROUGHPUT, AND MAC DELAY

	λ_f ($\sec^{-1}$)	10			15			20		
	λ_o (unit: λ_f)	10^{-2}	10^{-1}	10^{0}	10^{-2}	10^{-1}	10^{0}	10^{-2}	10^{-1}	10^{0}
Interfered Probability (p_c)	Analysis	0.001269	0.011963	0.069073	0.001711	0.015888	0.072858	0.002063	0.018935	0.065827
	Simulation	0.006657	0.011039	0.065376	0.007378	0.012342	0.070774	0.003893	0.011151	0.072634
	Error (%)	0.538817	0.09247	0.36973	0.566748	0.35459	0.20852	0.183074	0.77839	0.68076
Throughput (S)	Analysis	0.512132	0.59933	0.84713	0.517977	0.635336	0.874733	0.523682	0.665271	0.882073
	Simulation	0.522892	0.710338	0.734694	0.664444	0.77824	0.790637	0.748124	0.836423	0.830335
	Error (%)	1.076	11.1008	11.2436	14.6467	14.2904	8.4096	22.4442	17.1152	5.1738
MAC Delay ($E[T_M]$)	Analysis	0.005006	0.005078	0.0081	0.00501	0.00513	0.0114	0.005013	0.005193	0.01426
	Simulation	0.005091	0.005325	0.006027	0.005099	0.005365	0.005907	0.005058	0.005274	0.00571
	Error (%)	0.008479	0.024733	0.207267	0.008993	0.023499	0.549878	0.004447	0.008155	0.854896

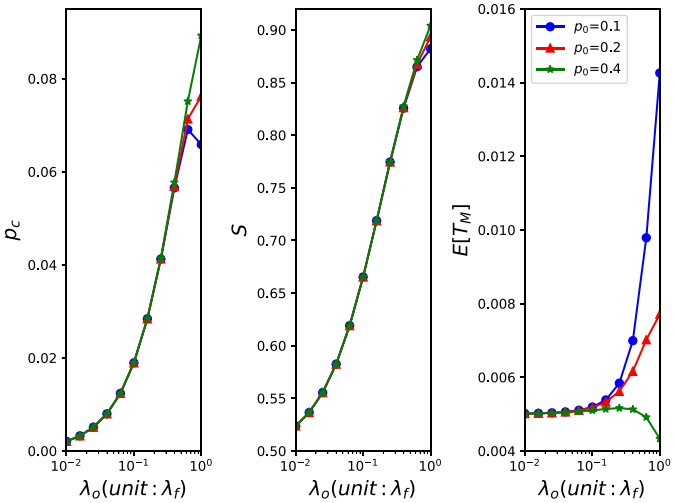


Fig. 6. Impact of p_0 on p_c, S, and $E[T_M]$ ($t_1 = 0.01$, $t_2 = 0.02$, $\lambda_f = 20$, $\gamma_m = 1/20$, and $\gamma_f = 1/20$). (a) Average interference probability of V2V. (b) Throughput of RSU. (c) Average MAC delay of RSU.

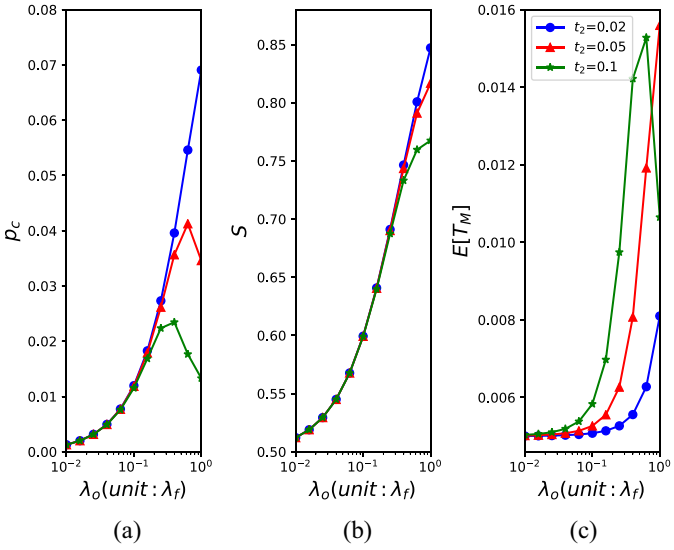


Fig. 7. Impact of t_2 on p_c, S, and $E[T_M]$ ($t_1 = 0.01$, $p_0 = 0.1$, $\lambda_f = 10$, $\gamma_m = 1/20$, and $\gamma_f = 1/20$). (a) Average interference probability of V2V. (b) Throughput of RSU. (c) Average MAC delay of RSU.

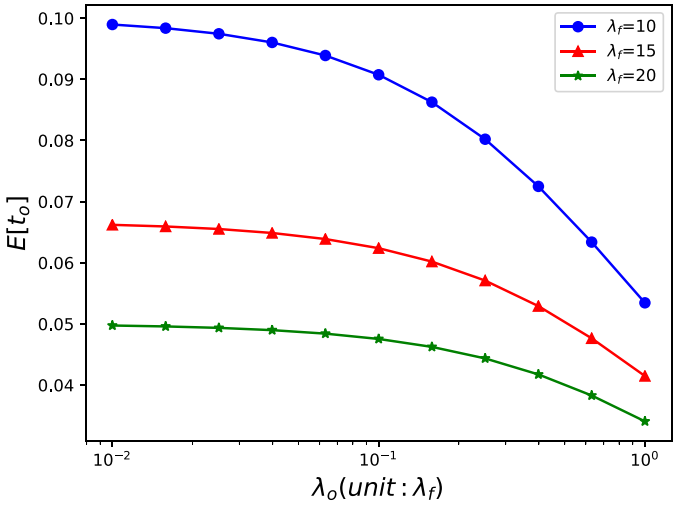


Fig. 8. Value of $E[t_o]$ with different value of λ_o, where $t_1 = 0.01$, $t_2 = 0.02$, $p_0 = 0.1$, $\gamma_m = 1/20$, and $\gamma_f = 1/20$.

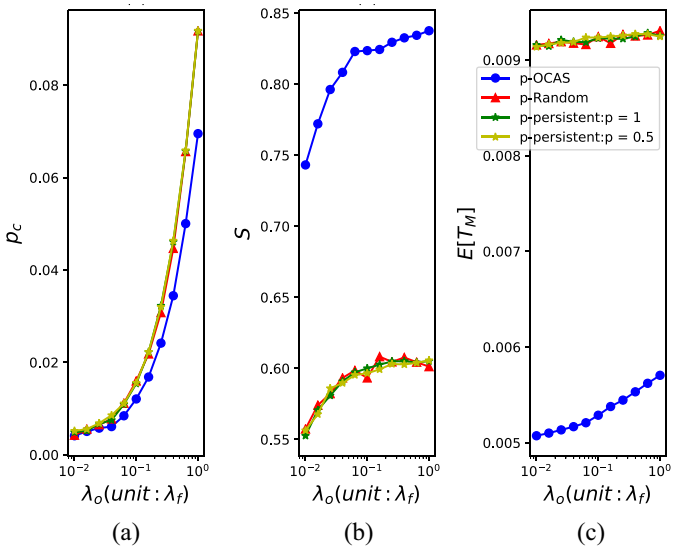


Fig. 9. Impact of λ_f on p_c, S, $E[T_M]$ ($t_1 = 0.01$, $t_2 = 0.02$, $p_0 = 0.2$, $\lambda_f = 20$, $\gamma_m = 1/20$, and $\gamma_f = 1/20$). (a) Average interference probability of V2V. (b) Throughput of RSU. (c) Average MAC delay of RSU.

half), high p_0 still results in high p_c than low p_0. However, the high p_0 can perform better S than low p_0 as shown in Fig. 6(b). This is because high p_0 enables RSUs to have higher probability to access the channel. After RSU defers for an appropriate period of time in order to avoid interfering the signals of a nearby AV, the probability of channel access will still be high. In this way, there are less unoccupied frames, and the

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

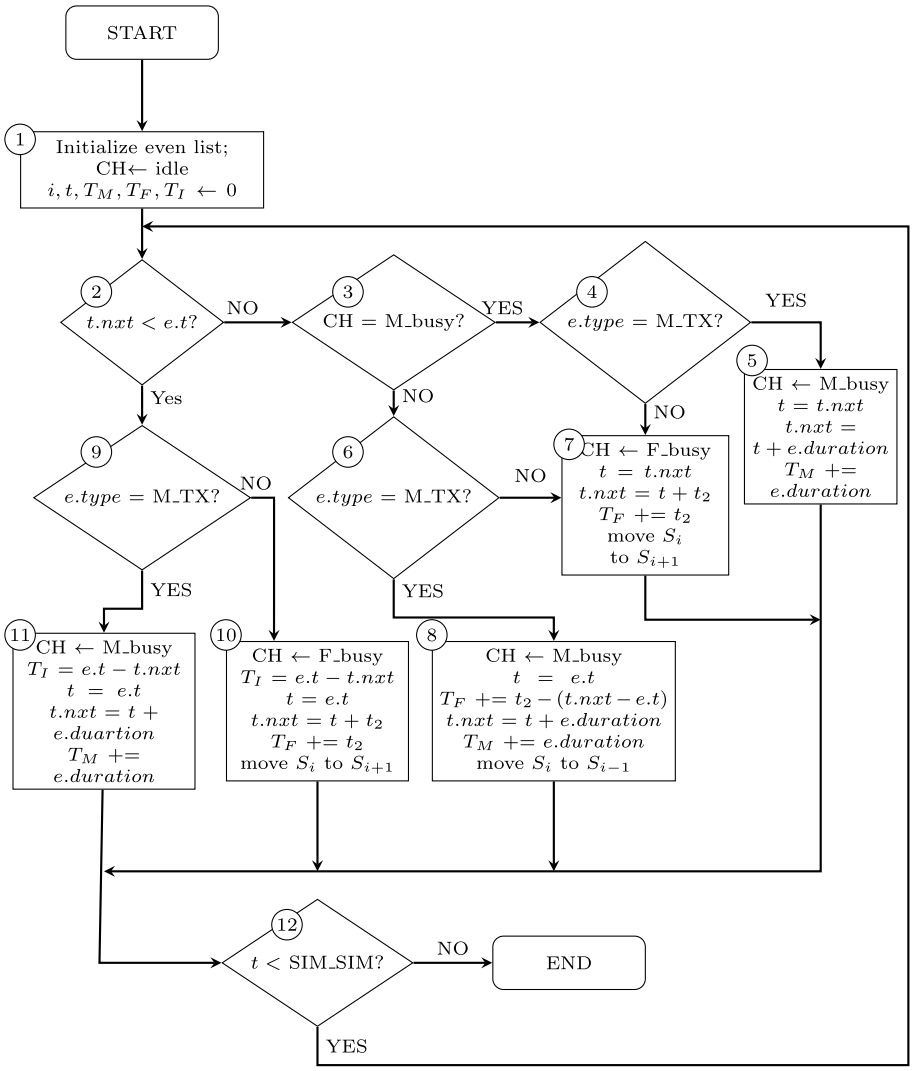


Fig. 10. Flowchart of *p*-OCAS.

channel utilization also becomes better. Fig. 6(c) shows that the time delay to achieve the state S_n under low p_0 is longer than that under high p_0 because the RSU with lower p_0 will need more times of consecutively successful channel access to reach S_n.

Fig. 7(a) illustrates that a higher value of t_2 will degrade the value of p_c especially when $\lambda_o = \lambda_f$. The reason is that a higher value of t_2 increases the probability that RSU turns its state from S_i to S_{i-1}, which decreases the channel access probability of RSU and mitigates the interference to the AVs. For example, the value of p_c under $t_2 = 0.02$ is about 4.8 times that under $t_2 = 0.01$ s when $\lambda_f = \lambda_o$. However, a higher t_2 makes the value of S become slightly lower when λ_o is close to λ_f as in Fig. 7(b). When $\lambda_f = \lambda_o$, the difference of S between $t_2 = 0.02$ s and $t_2 = 0.1$ s is about 8%. Fig. 7(c) shows that the delay of RSU is large with higher t_2. This is because RSU has high probability to encounter interference from AVs, which leads to RSU with low probability to access the channel. However, as the λ_o becomes large, the delay decreases in the case of higher t_2. This is because AV has more opportunity to channel attempt with very short waiting period.

Fig. 8 shows plotting of the value of $E[t_o]$ against different values of λ_o. It showed that with a fixed λ_o, the value of $E[t_o]$ under $\lambda_f = 10$ is larger than that under $\lambda_f = 15$ and $\lambda_f = 20$. This is because the AVs have more time to stay in the coverage of RSU, then it needs more time to delay to avoid interference. Meanwhile, it is obviously that $E[t_o]$ decreases as the value of λ_o increases. The reason is that the duration of the AV's V2V session becomes smaller when the λ_o becomes larger. Therefore, the value of $E[t_o]$ can be optimally obtained to improve the spectrum efficiency.

Fig. 9 presents the performance in p_c, S, and $E(T_M)$, compared with the *p*-random access scheme and *p*-persistent access scheme. For the *p*-persistent access scheme, the persistent probability p is set to 1 and 0.5. From Fig. 9, the difference between the *p*-random access scheme and the *p*-persistent access scheme is very small. However, the performance of the proposed *p*-OCAS outperforms both the *p*-random and *p*-persistent access schemes, where the average throughput S improves up to 36.8%, and the average MAC delay $E[T_M]$ and the interference probability p_c decreases up to 73.1% and 18.9%, respectively. This is because *p*-OCAS can dynamically adjust the channel access probability accordingly with

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

respect to the access load of the network at any point of time.

V. Conclusion

In this article, we addressed the co-channel problem in the 2-tier architecture of vehicular networks. A frame-based p-OCAS mechanism for RSUs in the 2-tier vehicular network was proposed to alleviate the problem of interference between V2V and V2I communication links. The effect of different parameters on the performance, such as the interference probability for V2V, the normalized throughput of RSU, and the average MAC delay of RSU had been critically investigated. It had been shown that the proposed p-OCAS can adaptively adopt the appropriate access probability according to the arrival rate of the AVs and thereby consequently reduces the chances of interference. The implementation cost of p-OCAS on RSU is limited because the mechanism only maintains a channel access probability p and is implemented in the MAC layer. This mechanism is cost effective because it is a software solution without any modification of the hardware. On the other hand, since p-OCAS complies the LTE specifications, it can be easily implemented into cellular chipsets and CAVs systems.

Appendix
Simulation Model

The Appendix describes the discrete-event simulation model for p-OCAS. Let T_M, T_F, and T_I denote the duration of AV access service, IU access service, and idle, respectively.

Step 1) Let the current state $i = 0$, channel access probability $f_p(i) = p_0$, and the number of all states $n = \lceil \log_2 p_0^{-1} \rceil$. Set the channel status be idle and $t, T_M, T_F, T_I = 0$.

Step 2) Get the newest event e from the event list, and check whether the start time of e, denoted as $e.t$, is later than the end time of the previous event, denoted as $t.nxt$. If yes, go to step 9. If no, go to step 3.

Step 3) Check whether the channel is occupied by an AV. If yes, go to step 4. Otherwise, go to step 6.

Step 4) When the channel is occupied by an AV, check the type of new event e, denoted as $e.type$, is M_TX. If yes, go to step 5. Otherwise, go to step 7.

Step 5) When the channel is occupied by an AV and $e.type =$ M_TX, the new event e is scheduled after the previous event. Set the simulation time t to the end time of the previous event, $t.nxt$, and set the end time of e to $t.nxt = t + e.duration$. To count the total duration of M_TX, denoted as T_M, add T_M by the duration of e, denoted as $e.duration$.

Step 6) When the channel is occupied by an RSU, check $e.type$ is M_TX. If yes, go to step 8. Otherwise, go to step 7.

Step 7) When the channel is occupied by an RSU and $e.type =$ F_TX, the new event e is scheduled after the previous event. Set the simulation time t to the end time of the previous event, $t.nxt$, and set the end time of e to $t.nxt = t + t_2$. To count the total duration of F_TX, denoted as T_F, add T_F by t_2, and move the state from S_n to S_{n+1}.

Step 8) When the channel is occupied by an RSU and $e.type =$ M_TX, the previous event is terminated and serve AV at $e.t$. Set $t.nxt = t + e.duration$, and increase T_M by $e.duration$. Because the transmission from an RSU to an IU is terminated, decrease the total duration of F_TX and move the state from S_n to S_{n-1}.

Step 9) If $e.t$ is later than $t.nxt$, it means that there is an idle period during $t.nxt$ to $e.t$. If $e.type$ is F_TX, go to step 10. Otherwise, go to step 11.

Step 10) When $e.type$ is F_TX, it will be served after the end of previous event that may be F_TX or M_TX. Set the channel to F_busy, and increase channel idle time, denoted as T_I, by $e.t - t.nxt$. Move t to $e.t$ and $t.nxt$ to $t + t_2$. After the FU is served by t_2, decrease $e.t$ by t_2 and move the state from S_n to S_{n+1}. If n does not reach the highest state $n = \lceil \log_2 p_0^{-1} \rceil$, move the state from S_i to S_{i+1}.

Step 11) When $t.nxt < e.t$ and $e.type$ are M_TX, the channel will be occupied by an AV. Set the channel state be M_busy, and move the current time to $e.t$. Then, count the idle time from $t.nxt$ to $e.t$, and count the T_M by $e.duration$.

Step 12) If the current time t is more than the simulation duration time T_SIM, presented as $t <$ T_SIM, the algorithm terminates and calculates the usage of IUs as $T_F/(T_F + T_M + T_I)$. Otherwise, go to step 2 and the algorithm continues running.

References

[1] "Intelligent transport systems (ITS); framework for public mobile networks in cooperative ITS (CITS), v1.1.1," ETSI, Sophia Antipolis, France, Rep. TR 102 962, Feb. 2012.

[2] H. Cheng, H. Shan, and W. Zhuang, "Infotainment and road safety service support in vehicular networking: From a communication perspective," *Mech. Syst. Signal Process.*, vol. 25, no. 6, pp. 2020–2038, Aug. 2011.

[3] K. M. Eshteiwi, K. B. Fredj, G. Kaddoum, and F. Gagnon, "Performance analysis of peer-to-peer V2V wireless communications in the presence of interference," in *Proc. IEEE PIMRC*, Montreal, QC, Canada, Oct. 2017, pp. 1–6.

[4] S. K. Kühlmorgen, H. Lu, A. Festag, J. Kenney, S. Gemsheim, and G. Fettweis, "Evaluation of congestion-enabled forwarding with mixed data traffic in vehicular communications," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 1, pp. 233–247, Jan. 2020.

[5] D. Elliott, W. Keen, and L. Miao, "Recent advances in connected and automated vehicles," *J. Traffic Transp. Eng.*, vol. 6, no. 2, pp. 109–131, Apr. 2019.

[6] B. Ko, H. Lee, and S. H. Son, "GPS-less localization system in vehicular networks using dedicated short range communication," in *Proc. 22nd Int. Conf. Embedded Real Time Comput. Syst. Appl. (RTCSA)*, Daegu, South Korea, Aug. 2016, p. 106.

[7] A. F. M. S. Shah and N. Mustari, "Modeling and performance analysis of the IEEE 802.11p enhanced distributed channel access function for vehicular network," in *Proc. Future Technol. Conf. (FTC)*, San Francisco, CA, USA, Dec. 2016, pp. 173–178.

[8] *IEEE Standard for Wireless Access in Vehicular Environments (WAVE)—Identifier Allocations*, IEEE Standard 1609.12-2016, pp. 1–21, Mar. 2016.

[9] 5GAA. (Sep. 2018). *V2X Technology Benchmark Testing*. [Online]. Available: https://www.fcc.gov/ecfs/filing/109271050222769

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.

[10] 5G Americas. (Mar. 2018). *Cellular V2X Communications Towards 5G*. [Online]. Available: https://www.5gamericas.org/cellular-v2x-communications-towards-5g

[11] X. Wang, S. Mao, and M. X. Gong, "An overview of 3GPP cellular vehicle-to-everything standards," *Mobile Comput. Commun.*, vol. 21, no. 3, pp. 19–25, Nov. 2017.

[12] G. Araniti, C. Campolo, M. Condoluci, A. Iera, and A. Molinaro, "LTE for vehicular networking: A survey," *IEEE Commun. Mag.*, vol. 51, no. 5, pp. 148–157, May 2013.

[13] H. Z. B. Sun, "IEEE 802.11-18/0861r9: 802.11 NGV proposed PAR," presented at the IEEE 802.11 NGV Meeting, Nov. 2019.

[14] "Study on enhancement of 3GPP support for 5G V2X services (v16.2.0, release 16)," 3GPP, Sophia Antipolis, France, Rep. TR 22.886, Dec. 2018.

[15] 3GPP, *New SID: Study on NR V2X*, document Plenary #RP-181480 Meeting, Vodafone, San Diego, CA, USA, pp. 1–10, Jun. 2018.

[16] U.S. Department of Transportation. (Jul. 2017). *Vehicle-to-Vehicle Communication Technology*. [Online]. Available: https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/v2v_fact_sheet_101414_v2a.pdf

[17] G. Naik, B. Choudhury, and J.-M. Park, "IEEE 802.11bd & 5G NR V2X: Evolution of radio access technologies for V2X communications," *IEEE Access*, vol. 7, pp. 70169–70184, 2019.

[18] A. Bazzi, G. Cecchini, A. Zanella, and B. M. Masini, "Study of the impact of PHY and MAC parameters in 3GPP C-V2V mode 4," *IEEE Access*, vol. 6, pp. 71685–71698, 2018.

[19] A. Bazzi, G. Cecchini, M. Menarini, B. M. Masini, and A. Zanella, "Survey and perspectives of vehicular Wi-Fi versus sidelink cellular-V2X in the 5G era," *Future Internet*, vol. 11, no. 6, p. 122, May 2019.

[20] Y. S. Song, S. K. Lee, J. W. Lee, D. W. Kang, and K. W. Min, "Analysis of adjacent channel interference using distribution function for V2X communication systems in the 5.9-GHz band for ITS," *ETRI J.*, vol. 41, no. 6, pp. 703–714, Sep. 2019.

[21] 5GAA. (Nov. 2016). *The Case for Cellular V2X for Safety and Cooperative Driving*. [Online] Available: https://5gaa.org/wp-content/uploads/2017/10/5GAA-whitepaper-23-Nov-2016.pdf

[22] S. Chen *et al.*, "Vehicle-to-everything (V2X) services supported by LTE-based systems and 5G," *IEEE Commun. Stand. Mag.*, vol. 1, no. 2, pp. 70–76, Jun. 2017.

[23] Q. Wang, P. Fan, and K. B. Letaief, "On the joint V2I and V2V scheduling for cooperative VANETs with network coding," in *Proc. 16th Int. Symp. Commun. Inf. Technol. (ISCIT)*, Ho Chi Minh, Vietnam, Jan. 2012, pp. 62–73.

[24] H. Ye, G. Y. Li, and B.-H. F. Juang, "Deep reinforcement learning based resource allocation for V2V communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 4, pp. 3163–3173, Apr. 2019.

[25] H. Yang, X. Xie, and M. Kadoch, "Intelligent resource management based on reinforcement learning for ultra-reliable and low-latency IoV communication networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4157–4169, May 2019.

[26] Y. Liu, H. Liu, and H. Chen, "Performance analyses of successive interference cancellation in vehicular network," in *Proc. 16th Int. Symp. Commun. Inf. Technol. (ISCIT)*, Qingdao, China, Sep. 2016, pp. 431–434.

[27] M. Yang, S.-W. Jeon, and D. K. Kim, "Interference management for in-band full-duplex vehicular access networks," *IEEE Trans. Veh. Technol.*, vol. 67, no. 2, pp. 1820–1824, Feb. 2018.

[28] A. Hisham, W. Sun, E. G. Ström, and F. Brännström, "Power control for broadcast V2V communications with adjacent carrier interference effects," in *Proc. IEEE ICC*, Kuala Lumpur, Malaysia, May 2016, pp. 1–6.

[29] Y. Ren, F. Liu, Z. Liu, C. Wang, and Y. Ji, "Power control in D2D-based vehicular communication networks," *IEEE Trans. Veh. Technol.*, vol. 64, no. 12, pp. 5547–5562, Dec. 2015.

[30] L. Liang, G. Y. Li, and W. Xu, "Resource allocation for D2D-enabled vehicular communications," *IEEE Trans. Commun.*, vol. 65, no. 7, pp. 3186–3197, Jul. 2017.

[31] J. Mei, K. Zheng, L. Zhao, Y. Teng, and X. Wang, "A latency and reliability guaranteed resource allocation scheme for LTE V2V communication systems," *IEEE Trans. Wireless Commun.*, vol. 17, no. 6, pp. 3850–3860, Jun. 2018.

[32] L. Liang, J. Kim, S. C. Jha, K. Sivanesan, and G. Y. Li, "Spectrum and power allocation for vehicular communications with delayed CSI feedback," *IEEE Wireless Commun. Lett.*, vol. 6, no. 4, pp. 458–461, Aug. 2017.

[33] L. Liang, S. Xie, G. Y. Li, Z. Ding, and X. Yu, "Graph-based radio resource management for vehicular networks," in *Proc. IEEE ICC*, Kansas City, MO, USA, May 2018, pp. 1–6.

[34] W. Sun, D. Yuan, E. G. Strom, and F. Brannström, "Cluster-based radio resource management for D2D-supported safety-critical V2X communications," *IEEE Trans. Wireless Commun.*, vol. 15, no. 4, pp. 2756–2769, Apr. 2016.

[35] M. Botsov, M. Klgel, W. Kellerer, and P. Fertl, "Location dependent resource allocation for mobile device-to-device communications," in *Proc. IEEE WCNC*, Istanbui, Turkey, Apr. 2014, pp. 1679–1684.

[36] L. Wu, Z. Zhang, J. Dang, and Y. Wu, "Frequency-domain intergroup interference coordination for V2V communications," *IEEE Signal Process. Lett.*, vol. 24, no. 11, pp. 1739–1743, Nov. 2017.

[37] R. Zhang, X. Cheng, Q. Yao, C. X. Wang, Y. Yang, and B. Jiao, "Interference graph-based resource-sharing schemes for vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 62, no. 8, pp. 4028–4039, Oct. 2013.

[38] F. Chiti, R. Fantacci, E. Dei, and Z. Han, "Context aware clustering in VANETs: A game theoretic perspective," in *Proc. IEEE ICC*, London, U.K., Jun. 2015, pp. 6584–6588.

[39] W. Saad, Z. Han, A. Hjorungnes, D. Niyato, and E. Hossain, "Coalition formation games for distributed cooperation among roadside units in vehicular networks," *IEEE J. Sel. Areas Commun.*, vol. 29, no. 1, pp. 48–60, Jan. 2011.

[40] T. Wang, L. Song, and Z. Han, "Coalitional graph games for popular content distribution in cognitive radio VANETs," *IEEE Trans. Veh. Technol.*, vol. 62, no. 8, pp. 4010–4019, Oct. 2013.

[41] M. I. Ashraf, M. Bennis, C. Perfecto, and W. Saad, "Dynamic proximity-aware resource allocation in vehicle-to-vehicle (V2V) communications," in *Proc. IEEE Globecom Workshops*, Washington, DC, USA, Dec. 2016, pp. 1–6.

[42] Y. Gu, L. X. Cai, M. Pan, L. Song, and Z. Han, "Exploiting the stable fixture matching game for content sharing in D2D-based LTE-V2X communications," in *Proc. IEEE GLOBECOM*, Washington, DC, USA, Dec. 2016, pp. 1–6.

[43] P. Wu, L. Ding, Y. Wang, and Y. Xiao, "Throughput analysis of vehicle-centric cooperative transmission in ultra-dense V2I networks," in *Proc. Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 3749–3754.

[44] H. Zhou *et al.*, "ChainCluster: Engineering a cooperative content distribution framework for highway vehicular communications," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 6, pp. 2644–2657, Dec. 2014.

[45] L. Kleinrock and F. Tobagi, "Packet switching in radio channels: Part I—Carrier sense multiple-access modes and their throughput-delay characteristics," *IEEE Trans. Commun.*, vol. COM-23, no. 12, pp. 1400–1416, Dec. 1975.

[46] J. Wang, Y. Shao, Y. Ge, and R. Yu, "A survey of vehicle to everything (V2X) testing," *Sensors*, vol. 19, no. 2, p. 334, Jan. 2019.

[47] 5GAA. (Dec. 2017). *An Assessment of LTE-V2X (PC5) and 802.11p Direct Communications Technologies for Improved Road Safety in the EU*. [Online] Available: http://5gaa.org/wp-content/uploads/2017/12/5GAA-Road-safety-FINAL2017–12-05.pdf

[48] Z. Xiao *et al.*, "Spectrum resource sharing in heterogeneous vehicular networks: A noncooperative game-theoretic approach with correlated equilibrium," *IEEE Trans. Veh. Technol.*, vol. 67, no. 10, pp. 9449–9458, Oct. 2018.

[49] *Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Channels and Modulation (Release 15), V15.1.0*, 3GPP Standard TS 36.211, Mar. 2018.

[50] R. W. Heath, M. Kountouris, and T. Bai, "Modeling heterogeneous network interference using Poisson point processes," *IEEE Trans. Signal Process.*, vol. 61, no. 16, pp. 4114–4126, Aug. 2013.

[51] Y. B. Lin, "Reducing location update cost in a PCS network," *IEEE/ACM Trans. Netw.*, vol. 5, no. 1, pp. 25–33, Feb. 1997.

[52] I. F. Akyildiz, J. S. M. Ho, and Y.-B. Lin, "Movement-based location update and selective paging for PCS networks," *IEEE/ACM Trans. Netw.*, vol. 4, no. 4, pp. 629–638, Aug. 1996.

[53] L. Kleinrock, *Queueing Systems: Theory, Volume I*. New York, NY, USA: Wiley, 1975.

[54] Y. Fang, I. Chlamtac, and Y.-B. Lin, "Call performance for a PCS network," *IEEE J. Sel. Areas Commun.*, vol. 15, no. 8, pp. 1568–1581, Oct. 1997.

[55] S. M. Ross, *Stochastic Processes*, 2nd ed. New York, NY, USA: Wiley, 1996.

[56] S. M. Ross, *Introduction to Probability Models*, 10th ed. Oxford, U.K.: Elsevier, 2010.

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.



Xiangming Wen (Senior Member, IEEE) received the B.E., M.S., and Ph.D. degrees in electrical engineering from Beijing University of Posts and Telecommunications (BUPT), Beijing, China.

He is currently the Vice President of BUPT, where he is also a Professor with the Communication Network Center and the Director of the Beijing Key Laboratory of Network System Architecture and Convergence. In the last five years, he has authored more than 100 published papers. His current research is focused on broadband mobile communication theory, multimedia communications, and information processing.

Prof. Wen is the Principle Investigator of more than 18 projects, including the National Key Project of Hi-Tech Research and Development Program of China (863 Program) and the National Natural Science Foundation of China. He is the Vice Director of the Organization Committee of the China Telecommunication Association.



Jenhui Chen (Senior Member, IEEE) received the B.S. and Ph.D. degrees in computer science and information engineering (CSIE) from Tamkang University, Taipei, Taiwan, in July 1998 and January 2003, respectively.

Since 2003, he has been with the Department of CSIE, School of Electrical and Computer Engineering, College of Engineering, Chang Gung University, Taoyuan, Taiwan, where he is currently a Full Professor and the Chairman. He serves, as the Chairman for the Bachelor Program in Artificial Intelligence (AI), Chang Gung University, where he is a Professor of the Center for AI in Medicine, Chang Gung Memorial Hospital. His main research interests include design, analysis, and implementation of communication protocols, wireless networks, AI, human-like intelligence, and data science.

Prof. Chen is currently a Senior Editor of *Cogent Engineering* and an Editor of the *Scientific World Journal* and *Advances in Computer Engineering*. He served as the Technical Program Committee Member for the IEEE Globecom from 2016 to 2019; IEEE VTC 2018; IEEE ICC 2015; IEEE ICCC from 2018 to 2019; IEEE ICCCN from 2014 to 2018; IEEE 5G World Forum from 2018 to 2019; and ACM CCIOT 2019. He also served as a reviewer for many well-known academic journals published by ACM, Elsevier, IEEE, and Springer.



Zhiqun Hu received the B.E. degree in communication engineering from Hubei University of Technology, Wuhan, China, in 2012, and the Ph.D. degree in information and communication engineering from Beijing University of Posts and Telecommunication, Beijing, China, in 2018.

She has been with Hubei University since 2018, where she is currently an Assistant Professor of computer science and information engineering. Her main research interests include wireless communications, deep reinforcement learning, performance analysis, V2X, and UAV communication.



Zhaoming Lu received the Ph.D. degree from Beijing University of Posts and Telecommunications, Beijing, China, in 2012.

He is currently an Associated Professor with the School of Information and Communication Engineering, Beijing University of Posts and Telecommunications. His research includes open wireless networks, QoE management in wireless networks, software-defined wireless networks, cross-layer design for mobile video applications, and network-assisted autonomous driving.

Authorized licensed use limited to: BIBLIOTECA DE LA UNIVERSIDAD DE CANTABRIA. Downloaded on September 01,2023 at 10:26:43 UTC from IEEE Xplore. Restrictions apply.